

THE PROBLEM

Indian banks lost **Rs 71,543 Cr** to fraud in FY 2024-25 (RBI Annual Report). Over 95% rely on rule-based AML with **95%+ false positive rates**, unable to detect multi-hop laundering (layering, round-tripping, structuring below CTR thresholds). STR preparation takes 4 to 6 hours manually. Industry average to identify fraud: **277 days** (IBM 2024).

Root cause: Rules check transactions in isolation. Sophisticated laundering is a network phenomenon. You need to see the graph.

THE SOLUTION

TRACE.ai models every transaction as a directed edge in a dynamic temporal graph (accounts = nodes, transactions = timestamped edges). Three detection layers run in parallel:

- Graph Pattern Matcher:** NetworkX algorithms detect circular flows, layering chains, structuring clusters, mule fan-in/fan-out, and dormant account bursts across 5 AML typologies.
- Temporal Graph Neural Network:** PyTorch Geometric TGN with edge-centric encoder (ChronoWave-GNN) and attention fusion (THG-OAFN). GraphSMOTE handles class imbalance. AUC greater than 96%.
- Online Anomaly Scorer:** River HalfSpaceTrees per-account baselines via streaming online learning. No batch retraining. ADWIN drift detection. Under 1ms per transaction.

INTELLIGENCE LAYER

- LLM Alert Explainer:** Human-readable alert explanation with full evidence chain.
- NL Investigation Copilot:** Plain-English queries translated to graph queries and returned as visual subgraphs. No Cypher required.
- Auto-STR Generator:** One-click FIU-IND compliant PDF with fund trail, risk breakdown, AI narrative. **4h to under 5 minutes.**

FIVE KEY INNOVATIONS

- 1.** Temporal GNN with edge-centric modeling (sequential transaction patterns)
- 2.** Streaming online ML: zero batch retraining + ADWIN drift detection
- 3.** Natural language investigation copilot (no technical query skills needed)
- 4.** Automated FIU-IND STR generation (4 hours reduced to 5 minutes)
- 5.** Hot-reloadable YAML compliance rules (instant regulatory updates)

BUSINESS MODEL

- B2B SaaS:** Annual licensing. Enterprise: Rs 1 to 2 Cr/yr. Mid-tier: Rs 50L to 1 Cr/yr. RRBs: Rs 10 to 30L/yr.
- SAM:** ~86 Indian banks at avg Rs 75L = Rs 64.5 Cr. SOM Year 1: Rs 2 to 5 Cr via Union Bank pilot + IBA endorsement.
- Vs. NICE Actimize/Oracle Mantas:** 10x cheaper, deploys in weeks, built India-native for RBI/FIU-IND from day one.

IMPACT AT A GLANCE

Rs 71,543 Cr Annual bank fraud (RBI FY 2024-25)	95%+ to <15% False positive rate reduction target
4h to 5 min STR preparation with TRACE.ai	277 days to Live Fraud identification speed
37% FP Reduction Temporal GNN vs. static baseline (2025)	>96% AUC GNN on comparable benchmarks

TECHNICAL ARCHITECTURE

Layer 0	Data Ingestion	CBS, NEFT/RTGS/UPI, IBM AMLSim
Layer 1	Graph Engine	NetworkX (proto), Neo4j + GDS (prod)
Layer 2	Detection	Pattern Matcher + TGN + Online ML + Rules + Fusion
Layer 3	Intelligence	LLM Explainer + NL Copilot + STR Generator
Layer 4	Delivery	FastAPI + React + 3D Force Graph + WebSocket

TECHNOLOGY STACK

Graph: NetworkX, Neo4j Community + GDS
GNN: PyTorch Geometric, TGN + ChronoWave encoder
Imbalance: GraphSMOTE + Focal Loss
Online ML: River HalfSpaceTrees + ADWIN
LLM: GPT-4o-mini / Claude Haiku via LiteLLM
NL Query: Vanna.ai-inspired NL to NetworkX / Cypher
Reports: ReportLab (FIU-IND STR PDFs)
Backend: FastAPI + WebSocket + SQLite
Frontend: React + react-force-graph-3d (Three.js)
Deploy: Docker Compose

RESEARCH FOUNDATION

- ChronoWave-GNN (2026): Edge-centric temporal GNN, SOTA on AMLSim
- THG-OAFN, PLOS ONE (2025): GraphSMOTE + attention fusion, AUC 96.56%
- TGN, Rossi et al., arXiv:2006.10637 (2020): Foundation temporal graph network
- Elliptic2 (2024): Subgraph laundering pattern validation
- BIS LVPS anomaly paper (2023): Two-layer detection framework
- Real-time TGNN for Payments (2025): 37% FP reduction benchmark
- RBI KYC Master Direction (2024) + PMLA 2002 + FIU-IND STR Manual